

Executive Summary

Project Overview

This project focuses on analyzing **customer churn** to understand the characteristics and service-related factors associated with customers leaving the company. The analysis was performed on a dataset containing **7,043 customers and 21 variables**, including demographic information, customer tenure, subscribed services, contract details, payment methods, monthly charges, total charges, and churn status.

The objective of the analysis was to identify **customer segments with a higher likelihood of churn**, understand the major patterns behind customer attrition, and generate actionable insights that could support customer-retention strategies.

Data Preparation and Quality Checks

Before performing the analysis, the dataset was inspected and cleaned to ensure that the results were reliable. The **TotalCharges** column was initially stored as a string and contained blank values. These blanks were replaced with **0** because they corresponded to customers with zero tenure and no recorded total charges, after which the column was converted into a numerical data type.

The dataset contained **no missing values after cleaning**, with all 7,043 records having valid observations across the 21 columns.

Duplicate checks were also performed. The analysis found **0 duplicate rows** and **0 duplicate customer IDs**, indicating that each customer record was unique in the dataset.

For better interpretability during visualization, the **SeniorCitizen** variable was also converted from a binary **0/1** representation into more readable **Yes/No** categories.

Overall Churn Situation

The churn analysis shows that **26.5% of the customers have churned**, while approximately **73.5% have remained with the company**. This means that roughly **1 out of every 4 customers** in the analyzed customer base has left the company.

This is an important business signal because customer churn directly affects recurring revenue, customer lifetime value, and the cost of acquiring replacement customers. Therefore, identifying the characteristics of customers who are more likely to churn is important for developing targeted retention strategies.

The churn distribution was visualized using both a **count plot** and a **percentage-based view**, making it easier to understand not only the absolute number of customers in each

group but also the overall proportion of the customer base represented by churned and retained customers.

Customer Tenure and Churn

One of the strongest patterns observed in the analysis is the relationship between **customer tenure and churn**.

Customers with **shorter tenure tend to show a substantially higher likelihood of churn**, particularly during the early stages of their relationship with the company. Customers who have only recently joined the service appear considerably more vulnerable to leaving compared with customers who have remained for a longer period.

The descriptive statistics show that the average customer tenure is approximately **32.4 months**, with a median of **29 months**. Tenure ranges from **0 to 72 months**, demonstrating a wide range of customer relationships in the dataset.

This suggests that the **early customer lifecycle is a critical retention period**. Rather than waiting until customers decide to leave, the company could focus retention efforts on new customers through onboarding, early engagement, service-quality checks, and proactive support.

Importantly, the analysis establishes an **association between shorter tenure and higher churn**, not that short tenure itself causes customers to churn.

Contract Type

Contract type is another major factor associated with customer churn.

Customers using **month-to-month contracts show substantially higher churn** compared with customers who have committed to one-year or two-year contracts. The analysis indicates that customers with longer contractual commitments are considerably more likely to remain with the company.

This pattern is particularly important from a business perspective because month-to-month customers have relatively low switching barriers. They can leave without the commitment associated with longer contracts.

Therefore, the company could consider strategies such as:

- Providing incentives for customers to move from month-to-month to annual contracts.
- Offering loyalty benefits for longer contractual commitments.
- Identifying month-to-month customers with high monthly charges as high-priority retention targets.
- Communicating the financial or service benefits of longer-term plans.

The key insight is that **contract structure appears to be strongly associated with customer retention**.

Internet Service and Churn

The analysis also highlights differences in churn across **InternetService** categories.

Customers using **Fiber optic service show comparatively higher churn** than customers using DSL or customers without internet service. This makes Fiber optic customers an important segment for further investigation.

However, this finding should not automatically be interpreted as meaning that Fiber optic service itself causes churn. A stronger business analysis would investigate whether other factors—such as **higher monthly charges, lack of support/security services, contract type, or payment method**—are concentrated within this customer group.

This is an important analytical distinction: the chart identifies a **high-risk segment**, while additional analysis is required to determine the underlying reason.

Additional Services and Customer Retention

The analysis of services such as **Online Security, Online Backup, Device Protection, and Tech Support** reveals another important pattern.

Customers who do **not** subscribe to these additional services generally show higher churn compared with customers who use them.

This may indicate that customers with additional services have greater engagement with the company's ecosystem and therefore stronger reasons to remain customers. Alternatively, customers who purchase multiple services may already be more committed to the company.

From a business perspective, these services could therefore represent an opportunity for both **customer engagement and retention**.

Potential strategies include:

- Bundling additional services with core plans.
- Offering introductory discounts on security and support services.
- Educating customers about the value of these services.
- Targeting high-risk customers with relevant service upgrades.

Again, these relationships should be interpreted as **associations rather than causal relationships**.

Payment Method

Payment method is another noticeable churn indicator.

Customers using **Electronic Check** demonstrate comparatively higher churn than customers using automatic payment methods such as **Bank Transfer or Credit Card**.

This pattern could indicate that customers using manual payment methods have different engagement or billing behaviors. It may also indicate that automatic payment customers are more likely to maintain their subscriptions because payments occur without requiring recurring manual action.

From a retention perspective, the company could encourage customers to adopt **automatic payment methods** through simple enrollment processes or appropriate incentives.

However, payment method should be treated as a **risk indicator rather than proof of a cause of churn**. Further investigation into billing issues, payment failures, customer preferences, and pricing would be required before implementing a policy based on this finding.

Customer Charges

The dataset also provides useful information about customer spending.

The average **monthly charge is approximately \$64.76**, while the median monthly charge is approximately **\$70.35**. Monthly charges range from **\$18.25 to \$118.75**.

Average total charges are approximately **\$2,279.73**, with a median of approximately **\$1,394.55**. Total charges range from **\$0 to \$8,684.80**.

The difference between the mean and median of total charges indicates substantial variation in the amount customers have spent with the company. This is important because losing a long-tenure, high-value customer can have a considerably different financial impact from losing a recently acquired customer.

Therefore, churn prevention should ideally consider **both churn probability and customer value**.

Key Customer Segments at Risk

Combining the patterns identified throughout the analysis, the customer groups that deserve particular attention include:

1. **New customers with short tenure**
2. **Month-to-month contract customers**
3. **Fiber optic customers**
4. **Customers without Online Security**
5. **Customers without Online Backup**
6. **Customers without Device Protection**
7. **Customers without Tech Support**
8. **Customers using Electronic Check**
9. **Customers with combinations of several of the above characteristics**

The last point is particularly important. A customer may not be high-risk because of one characteristic alone. For example, a **new customer on a month-to-month contract using**

Fiber optic service and Electronic Check may represent a much more important retention opportunity than a customer who possesses only one of these characteristics.

Business Recommendations

Based on the patterns identified in the analysis, the company should consider a **targeted retention strategy rather than treating every customer equally**.

1. Focus on early-stage customers

The first few months appear to be an important period for churn. The company should strengthen onboarding, engagement, and proactive customer support during this stage.

2. Encourage longer-term contracts

Month-to-month customers represent an important churn-risk segment. Appropriate incentives could encourage customers to move toward one-year or two-year contracts.

3. Investigate Fiber optic churn

Since Fiber optic customers demonstrate comparatively higher churn, the company should investigate whether the issue is related to pricing, service quality, technical support, customer expectations, or the combination of other customer characteristics.

4. Promote value-added services

Customers without services such as Online Security, Online Backup, Device Protection, and Tech Support show higher churn tendencies. Carefully designed bundles and targeted offers could increase customer engagement.

5. Encourage automatic payments

The higher churn observed among Electronic Check users suggests that payment-method behavior deserves attention. Encouraging automatic payment options may improve payment convenience and customer retention.

6. Prioritize customers based on value

Retention efforts should not focus only on the probability of churn. Customers should also be evaluated based on **monthly charges, tenure, and total customer value** so that limited retention resources are directed toward the customers where they can have the greatest business impact.

Overall Conclusion

The analysis demonstrates that customer churn is **not evenly distributed across the customer base**. Instead, churn is associated with identifiable customer and service

characteristics. The strongest patterns point toward **short-tenure customers, month-to-month contracts, Fiber optic users, customers without additional support/security services, and Electronic Check users.**

With an overall churn rate of approximately **26.5%**, the business has a meaningful opportunity to improve retention by identifying high-risk customers early and applying targeted interventions.

The analysis also demonstrates the importance of moving beyond simple churn counts. **Percentages and segment-level comparisons provide a clearer picture of where churn is concentrated**, while combining tenure, contract, service, payment, and financial characteristics can help the business move toward a more effective customer-retention strategy.

Overall, this project provides a foundation for a **data-driven churn-management framework**, where customer behavior is analyzed to identify risk segments, prioritize retention actions, and ultimately improve customer lifetime value.